

Conditional Probability

Let's start/review with an example with... wait for it... a pair of 6-sided dice!

Example:

A pair of 6-sided dice is rolled. Let x represent if the roll is doubles. Let y represent the sum on both dice. Then:

$x \in \{0,1\}$ (where 0 corresponds to a non-doubles roll, and 1 corresponds to a doubles roll)

$y \in \{2,3,4,5,6,7,8,9,10,11,12\}$ (the possible sums)

Find the probability of rolling doubles given that the roll was a sum of 4.

First of all, we note that we can easily solve this directly by considering the event space given that a sum of 4 was rolled; it consists of the following three rolls: $\{(1,3),(2,2),(3,1)\}$. Only one of these, (2,2), is

doubles, so we have that $P(\text{doubles} \mid \text{sum of } 4) = P(x = 1 \mid y = 4) = \frac{1}{3}$.

Alternatively, we can solve this indirectly using Bayes' Theorem: $P(x \mid y) = \frac{P(x) \cdot P(y \mid x)}{P(y)}$

$$\begin{aligned} P(x = 1 \mid y = 4) &= \frac{P(x = 1) \cdot P(y = 4 \mid x = 1)}{P(y = 4)} \\ &= \frac{\frac{6}{36} \cdot \frac{1}{6}}{\frac{3}{36}} \\ &= \frac{1}{3} \end{aligned}$$

So the probability of rolling doubles given that the roll was a sum of 4 is about 0.3333.

Conditional Probability in Cryptosystems

For a cryptosystem (M, C, K, E, D) where:

$x \in M$, the set of all plaintext strings

$y \in C$, the set of all ciphertext strings

$k \in K$, the set of all encryption keys

$e_k \in E$, the set of all encryption rules for a key k (every rule can be the same encryption function)

$d_k \in D$, the set of all decryption rules for a key k (every rule can be the same decryption function)

Consider the problem of finding the probability that the plaintext was “info” given that the ciphertext is “33921”.

To solve this problem using Bayes’ theorem we need to know:

$P(x = \text{info})$, the probability of this particular plaintext being encrypted; for example we would expect that the plaintext string “info” would be more probable than say “zxhq”.

$P(y = 33921 | x = \text{info})$, the probability of the cipher text being 33921 given that the plaintext is “info”; this depends on knowing all the possible ways of obtaining a ciphertext of 33921 using the all the keys k in K , and if some keys are more likely to be used than others.

$P(y = 33921)$, the overall probability of the cipher text being 33921; again this depends on knowing all the possible ways of obtaining a ciphertext of 33921 using the all the keys k in K , and if some keys are more likely to be used than others.

Anyone who has an encryption matrix for the cryptosystem and the probability distributions for both x and k can solve this using Bayes’ theorem:

$$P(x = \text{info} | y = 33921) = \frac{P(x = \text{info}) \cdot P(y = 33921 | x = \text{info})}{P(y = 33921)}$$

Example:

A cryptosystem consists of the following plaintext, ciphertext, and key sets:

$x \in M = \{a, b\}$, the set of all plaintext strings

$y \in C = \{1, 2, 3, 4\}$, the set of all ciphertext strings

$k \in K = \{k_1, k_2, k_3\}$, the set of all encryption keys

The probability distributions for x and k are: $P(a) = \frac{1}{4}, P(b) = \frac{3}{4}$ and $P(k_1) = \frac{1}{2}, P(k_2) = \frac{1}{4}, P(k_3) = \frac{1}{4}$

The encryption matrix is:

	a	b
k_1	1	2
k_2	2	3
k_3	3	4

Find the probability that the plaintext was " b " given that the ciphertext is "2".

$$P(x = b) = \frac{3}{4}$$

$$P(y = 2 | x = b) = P(k_1) = \frac{1}{2}$$

$$\begin{aligned}
 P(y = 2) &= P(k_1) \times P(y = 2 | k_1) + P(k_2) \times P(y = 2 | k_2) \\
 &= \frac{1}{2} \times \frac{3}{4} + \frac{1}{4} \times \frac{1}{4} \\
 &= \frac{3}{8} + \frac{1}{16} \\
 &= \frac{7}{16}
 \end{aligned}$$

$$P(x = b | y = 2) = \frac{P(x = b) \cdot P(y = 2 | x = b)}{P(y = 2)} = \frac{\frac{3}{4} \cdot \frac{1}{2}}{\frac{7}{16}} = \frac{\frac{3}{8}}{\frac{7}{16}} = \frac{3}{8} \times \frac{16}{7} = \frac{6}{7}$$

So the probability that the plaintext was " b " given that the ciphertext is "2" is about 0.8571.

Exercises:

1. A pair of 6-sided dice is rolled. Let x represent if the roll is doubles (1 for doubles, 0 otherwise) and let y represent the sum on both dice. Find:

- a) $P(x = 1 | y = 6)$
- b) $P(y = 6 | x = 1)$
- c) $P(x = 0 | y = 5)$
- d) $P(x = 0 | y = 2)$
- e) $P(x = 1 | y \leq 4)$

2. A cryptosystem has encryption matrix:

	a	b
k_1	1	2
k_2	2	3
k_3	3	4

And probability distributions: $P(a) = \frac{1}{4}, P(b) = \frac{3}{4}$ and $P(k_1) = \frac{1}{2}, P(k_2) = \frac{1}{4}, P(k_3) = \frac{1}{4}$

Find the probability that:

- a) the plaintext was " a " given that the ciphertext is "2".
 - b) the plaintext was " a " given that the ciphertext is "3".
 - c) the plaintext was " b " given that the ciphertext is "1".
 - d) the plaintext was " b " given that the ciphertext is "3".
3. If the chance of you being hacked is 20%, the chance of your network being vulnerable is 10%, and the chance of your network being vulnerable given that you have been hacked is 40%, what is the chance that you will be hacked given that that your network is vulnerable?

Answers:

1.

- a) 0.2000
- b) 0.1667
- c) 1.0000
- d) 0.0000
- e) 0.3333

2.

- a) 0.1429
- b) 0.2500
- c) 0.0000
- d) 0.7500

3. 0.8000